Synergetic Focal Loss for Imbalanced Classification in Federated XGBoost

Jiao Tian, Pei-Wei Tsai, *Member, IEEE*, Kai Zhang, Xinyi Cai, Hongwang Xiao, Ke Yu, Wenyu Zhao, and Jinjun Chen, *Fellow, IEEE*

***Abstract*—Applying sparsity- and overfitting-aware eXtreme Gradient Boosting (XGBoost) for classification in federated learning allows many participants to train a series of trees collaboratively. Since various local multiclass distributions and global aggregation diversity, model performance plummets as convergence slowly and accuracy decreases. Worse still, neither the participants nor the server can detect this problem and make timely adjustments. In this article, we provide a new local-global class imbalance inconsistency quantification and utilize softmax as the activation and focal loss, a dynamically scaled cross-entropy loss, in federated XGBoost to mitigate local class imbalance. Moreover, we propose a simple but effective hyperparameter determination strategy based on local data distribution to adjust the sample weights among noncommunicating participants, synergetic focal loss, to solve the inconsistency of local and global class imbalance, a unique characteristic of federated learning. This strategy is perfectly integrated into the original classification algorithm. It requires no additional detectors or information transmission. Furthermore, a dynamical for loop is designed to capture an optimum hyperparameter combination. Finally, we conduct comprehensive tabular- and image-based experiments to show that synergetic focal loss used in federated XGBoost achieves faster convergency and significant accuracy improvement. Simulation results prove the effectiveness of the proposed principle of configuring sample weights.**

***Impact Statement*—With the rapidly growing daily data and the increasing awareness of customers' privacy, the public applies Federated Learning to solve the data island problem in model training. Deep learning has relatively high requirements on edge computing devices, and XGBoost, which has performed well in both industrial practice and Kaggle competition, becomes a good choice. However, as many clients participate in Federated Learning, statistical heterogeneity occurs due to global and local class imbalance. Combined with focal loss, our proposed new algorithm alleviates the above phenomena through comprehensive scenario experiments. In addition, we show entirely that higher-accuracy loop algorithms are not generally needed. This technical solution can be applied to all federated XGBoost multi-classification tasks on various data types.**

***Index Terms*—Federated learning, focal loss, nonindependent and identical distribution (non-IID), XGBoost.**

Manuscript received 27 October 2022; revised 25 January 2023 and 4 March 2023; accepted 4 March 2023. Date of publication 8 March 2023; date of current version 12 February 2024. This paper was recommended for publication by Associate Editor X. Li upon evaluation of the reviewers' comments. *(Corresponding author: Jiao Tian.)*

Jiao Tian, Pei-Wei Tsai, Kai Zhang, Xinyi Cai, Wenyu Zhao, and Jinjun Chen are with the Department of Computing Technologies, Swinburne University of Technology, Melbourne, VIC 3122, Australia (e-mail: tianjiao8886@gmail.com; ptsai@swin.edu.au; kevin.zhang0522@gmail.com; xcai@swin.edu.au; wenyuzhao1993@hotmail.com; jinjun.chen@gmail.com).

Hongwang Xiao is with the Center for Health Computing Research, Beijing Academy of Artificial Intelligence, Beijing 100084, China (e-mail: hwxiao@baai.ac.cn).

Ke Yu is with the School of Big Data and Software Engineering, Chongqing University, Chongqing 400044, China (e-mail: keyu@cqu.edu.cn).

Digital Object Identifier 10.1109/TAI.2023.3254519

I. Introduction

FEDERATED learning (FL) [1] has become a popular distributed framework that enables tremendous participants who own local datasets cooperatively to train a joint machine learning (ML) model. FL essentially solves problems of client privacy [2], data and profit ownership [3], and takes advantage of edge computation [4] to a large extent. Classification methods play a crucial role in knowledge discovery of databases and data mining, such as sparse support vector machine [5], stochastic gradient descent (SGD) based deep learning [6], graph neural networks [7], gradient boosting decision trees (GBDTs) [8], and meta-learning [9] is widely used in FL. XGboost [10], with highly efficient, flexible, and portable characteristics, is famous for both classification and regression tasks. It was utilized in many Kaggle competitions winning teams and was the third most commonly used ML algorithm. In the foreseeable future, eXtreme Gradient Boosting (XGBoost) and its variants are still the most popular methods in the data science community.

We mainly focus on the multiclass imbalanced problem in federated XGBoost, which increases the severity and complexity of class imbalance situations and rules out many existing well-developed binary classification methods. In standard horizontal FL [11], each participant has nearly similar features but might have totally different samples, which will show inconsistencies in the distribution of multiple classes within each node while all participants are from the same domain. Moreover, the aggregated distribution of all data also performs various widely. We introduced local and global imbalance as the abovementioned two types of class imbalance in FL [12]. In general, such inconsistency could be pretty significant in practice even in the situation that a minority global class occupies the overwhelming majority of data on a particular participant. These are all manifestations of the nonindependent and identical distribution (non-IID) label problem. However, there are no standard measures of global imbalance caused by superimposing local imbalance effect in FL and local global imbalance inconsistency caused by this bias at present. Through both empirical and mathematical analysis, we observe the L2 norm of the data distribution to measure this FL specific phenomenon is appropriate. Furthermore, many methods that work well in binary classification are unavailable or perform mediocrely in multiclassification since the relationships among classes are no longer straightforward, and the boundaries among classes might overlap and are hard to distinguish immediately [13]. From the privacy perspective, FL only communicates gradients and the common model between the server and participants [14]. Additional information about local data is best left unpassed, making existing technologies hard to assemble a data analysis and a training model that fits the characteristics of globally aggregated data. No matter the rise in complexity among classes, the interweaving of global and local imbalance problems, or the consideration of privacy protection, the original model's convergence speed and prediction accuracy will be affected.

Considering the previous difficulties, federated XGBoost with stochastic distributed local multiple classes requires more modulation. This article utilizes focal loss to simulate the process of separating the well-classified samples and putting them aside, then focusing on the hard-classified samples that are consistently misclassified. By the way, the focal loss can calibrate the local imbalance problem because most ML generally predicts minority classes into majority classes to obtain relatively better overall accuracy. The focal loss will slightly mitigate class imbalance by giving those errors an insubstantial weight decay. However, local-global imbalance inconsistency needs dynamically adjust sample weights to cooperate with the focal loss so that the data distribution among participants could be close and achieve adjustment of global class imbalance simultaneously. We propose an easy but efficient hyperparameter determination strategy, synergetic focal loss. Each participant modifies sample weights based on local class distribution after receiving a certain hyperparameter without communicating extra data information with the server, resulting in an aggregated update that also mitigates global class imbalance. Simultaneously, it also cleverly avoids the model failure caused by the repeated superposition of sample augmentation without a mediator. Initially, selecting two hyperparameters in focal loss is easy by accumulating experience in binary classification. While it becomes exponentially time, communication, and computational consuming with multiple participants and multiclassification in federated XGBoost. According to our strategy, professional experimental experience is not required to adjust the combination of parameters among the FL participants.

Our contributions are as follows.

1) We implement focal loss to XGBoost to address the multiclass imbalanced problem and conduct detailed derivation in an FL environment.
2) We introduced a new quantifying algorithm to measure the degree of the inconsistency of global and local imbalance in FL.
3) An ingenious hyperparameter determination strategy, synergetic focal loss, is proposed. This method mitigates global and local imbalances in federated XGBoost and the inconsistency between the two types of imbalances.
4) Comprehensive experiments are provided to demonstrate the improvement of performance and convergence of synergetic focal loss in federated XGBoost training on tabular- and image-based projects.

II. Related Work

This article builds on a long line of works regarding FL, XGBoost, and class imbalance learning. As far as our contributions are concerned proposing a new distributed class imbalance quantification and a class imbalance mitigation in federated XGBoost, we review the previous literature on three topics separately.

A. Federated Learning

A standard FL objective function is the weighted average of each participant's local loss function. The most common form of aggregation is called FedAvg [1], including using native local optimizers such as SGD, RMSProp, and related variants that have been analyzed in the IID environment [15], [16]. The learning rate must decay on non-IID data for FedAvg with parallel SGD [17]. Simultaneously, because of the characteristics of FedAvg that are not well tuned and unfavorable for convergence, adaptive optimizers of federated version, including SGDm, RMSProp Adagrad, Adam, and Yogi, are proposed and their convergence and communication efficiency with nonconvex settings in non-IID environments are analyzed [18], [19]. Another way to solve non-IID is to modify the aggregation function. FedProx was proposed to leverage the proximal SGD to make local updates close to the global model [20]. FedNova subsumed FedAvg, FedProx, and Momentum SGD used in local participants to generate a normalized averaging formulation [21]. Subsequently, some methods to add participants-assisted drift correction terms to the loss function also appeared [22], [23].

B. XGBoost

Most literature shadow light on neural network-based FL, and tree-based FL models are rarely studied because of the need to overcome sequential additive and iterative characteristics during tree generation. With shortages of easy-to-overfit in small datasets for the light gradient boosting machine [24] and time-consuming for Catboost [25], XGBoost, an advanced gradient boosting decision tree-based algorithm, fits the requirements of efficiency, scalable, limit demand for hardware and robust lightweight model in an FL framework. XGBoost is widely applied in research and application scenarios, such as disease diagnostics and analysis in the medical industry; classification, prediction in financial default behaviors, IoT, and so on. Amazon SageMaker, a leader in cloud computing, has deployed XGBoost models for customers to do general ML predictions with a few simple buttons. Facebook launched its GBDT+ logistics regression AD recommendation system in 2014. Additionally, iQiyi designed and developed its flexible and high-performance XGBoost Serving reasoning system for its own production environment in 2021.

C. Class-Imbalanced Learning

The widespread application deployment of artificial intelligence in ubiquitous daily life has sparked interest in leveraging private data that often occur class imbalance. For example, credit defaults account for a very small percentage of total trading volume but can cause huge economic losses. The Industrial and Commercial Bank of China has launched a bond default risk calculation FL model based on the deposited corporate funds, repayment ability, and the bond risk label data of its subsidiary ICBC Credit Suisse. Dignity Health, one of America's largest healthcare systems, connects 39 hospitals, and more than 900 institutions to share data to prevent, predict, and treat diseases like sepsis, kidney failure, and so on. Centralized machine learning usually gives the solution of data-level and algorithm-level for class-imbalance problems, and the former can be divided into oversampling and undersampling. Participants should not arbitrarily oversampling or undersampling in FL without considering the distribution of other participants' data. This will pile up damage to data information. Several mitigations and calibrations have been put forward in the literature for the class imbalance problem in FL. Data augmentation, such as vanilla, mixup [26], and the generative adversarial network [27], achieved through sharing data information includes a general global dataset or aggregated participants' data distribution information [28], [29]. Mediators, a multiaimed combination scheme, and deep Q-learning were proposed to select the client set to achieve partial equilibrium [19], [30], [31]. Abovementioned data-level solutions leak more or less raw information to achieve global data balance. Moreover, it is infeasible for FL participants to learn tail classes' information through communication with other participants in global imbalanced scenarios [32]. With heterogeneous data shards, algorithm-level powerful personalization is another branch [33] and the dynamic modified sharing model is more suitable without more privacy leakage, which can be separated as local optimization regularization [31], [34], [22] and model aggregation scheme improvement [35], [36] for neural network and deep learning. In terms of XGBoost in FL, it can handle non-IID data to a certain extent [37]. There is no further improvement for XGBoost class imbalance in FL at present.

III. Problem Setup

In this section, we exhibit how XGBoost is applied in the FL framework. Then, we proposed a data distribution-based class imbalance quantification to assist the experimental setting.

A. Preliminary of Federated XGBoost

XGBoost is a highly efficient, scalable additive learning scheme through second-order Taylor expansion approximation of loss function. XGBoost objective for trees ensemble with regularization is written as

$$Obj^{(t)} = \sum_{m=1}^{M}\sum_{i=1}^{N_m} l\left(y_m^i,\ \hat{y}_m^{i(t-1)} + f_t\left(x_m^i\right)\right) + \Omega\left(f_t\right) \tag{1}$$

where M is the number of participants in FL, each participant, P_m, has $N_m(m \in [1, M])$ samples. $\{(y_m^i|x_m^i)\}$ denotes as the data in P_m, x_m^i is the feature vector from *i*th instance of P_m, y_m^i represents the multiple class label, $l(\cdots, \cdots)$ is the loss function and $\Omega(f_t)$ is the regular term, which can be expressed as $\Omega\left(f_t\right) = \gamma T + \lambda \sum_{j=1}^{T} w_j^2/2$ that contain two punishments, number of leaves T and L2 norm of leaf scores w_j. The additive manner based on the residual subtracted last predicted tree results is described as $\hat{y}_m^{i(t)} = \hat{y}_m^{i(t-1)} + f_t(x_m^i)$, where t is the number of iterations during the training process. Considering the first and second derivatives of the objective function, (1) can be converted to

$$Obj^{(t)} = \sum_{m=1}^{M}\sum_{i=1}^{N_m}\left[g_m^i f_t\left(x_m^i\right) + \frac{1}{2}h_m^i\left(f_t\left(x_m^i\right)\right)^2\right] + \Omega\left(f_t\right) \tag{2}$$

where $g_m^i = \partial_{\hat{y}_m^{i(t-1)}} l(y_m^i,\ \hat{y}_m^{i(t-1)})$ and $h_m^i = \partial^2_{\hat{y}_m^{i(t-1)}} l(y_m^i,\ \hat{y}_m^{i(t-1)})$ are gradient and hessian.

Defining a new additive tree by a vector of scores in leaves, then $f_t\left(x_m^i\right) = w_{q(x_m^i)}$, $w \in R^T$, $q : R^{d_m} \to \{1, 2, \cdots, T\}$. q is the transformation from example to tree leaves, w is the corresponding weight. Finally, optimal weight w_j^* of leaf j is

$$w_j^* = -\frac{\sum_{m=1}^{M} G_m^j}{\sum_{m=1}^{M} H_m^j + \lambda} \tag{3}$$

where G_m^j is the sum of g_m^i that belong to a certain leaf $I_j = \{i|q(x_m^i) = j\}$. Similarly, we have H_m^j. Putting the optimal weight into the objective function and building a new tree greedily, we obtain the tree-splitting criterion straightforward

$$\mathrm{Max}\left([A + B - C]\right) \tag{4}$$

where $A = \left(\sum_{m=1}^{M} G_m^L\right)^2 \Big/ \left(\sum_{m=1}^{M} H_m^L + \lambda\right)$, $B = \left(\sum_{m=1}^{M} G_m^R\right)^2 \Big/ \left(\sum_{m=1}^{M} H_m^R + \lambda\right)$, $C = \left(\sum_{m=1}^{M} G_m^L + \sum_{m=1}^{M} G_m^R\right)^2 \Big/ \left(\sum_{m=1}^{M} H_m^L \sum_{m=1}^{M} H_m^R + \lambda\right)$ that $G_m^L = \sum_{i \in I_L} g_m^i$ and $G_m^R = \sum_{i \in I_R} g_m^i$ represent the sum of gradients in left and right bifurcation nodes in the new splitting tree. The same goes for hessian.

B. Global and Local Class Imbalance Quantification

The federated XGBoost does not tackle the class imbalance problem, one of the widespread non-IID in horizontal FL, and is also named label distribution skew [38]. The Astrea framework rebalances the participants' dataset by augmenting the raw dataset, which will spend more computation that profoundly impacts FL efficiency [39]. Data resampling utilizes an adequate number of samples [40] to rebalance the sample weight, which benefits from the volume of samples provided in [32]. While the volume of samples in XGBoost has no such relationship with learning benefit since it is an ensemble additive model. Fed-focal loss is extended into multilayer perceptron for image-based imbalanced data classification [34].

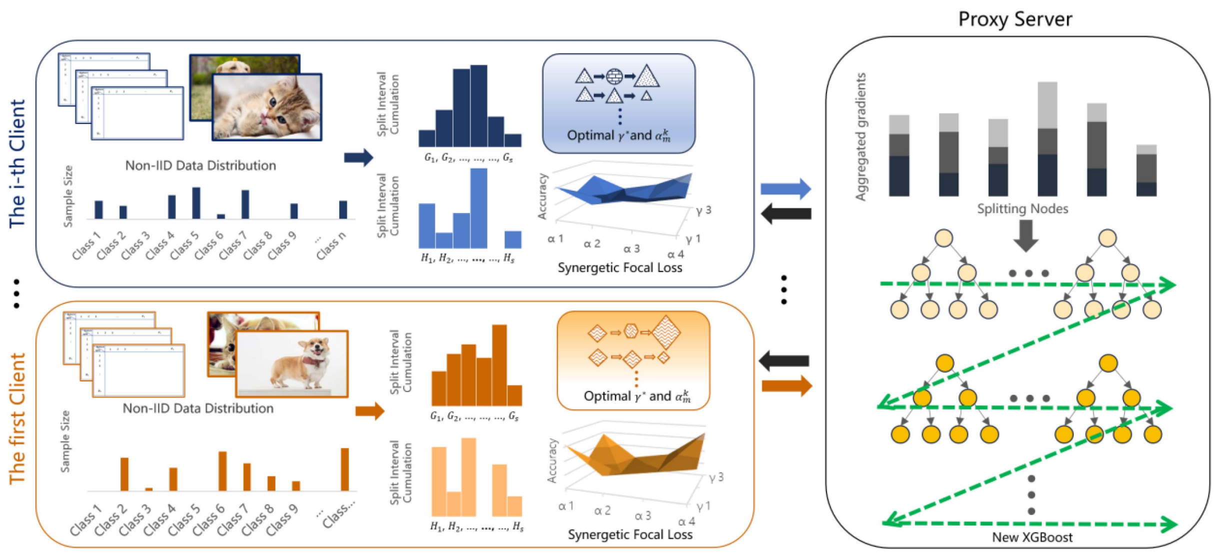

Fig. 1. Flowchart of synergetic focal loss in federated XGBoost.

Particularly with respect to FL, the traditional global class imbalance is accompanied by a new problem: local class imbalance training based on a local raw dataset will accumulate bias with the aggregation paradigm. In detail, the number of samples for class k denoted by N_m^k on every local participant P_m, K is the number of whole classes. Local class imbalance percentage ρ_m for device P_m is defined as $\rho_m = N_m^1 : N_m^2 : \cdots : N_m^k$ if this is not even distribution. We use various percentage $\rho_{global} = \sum_{m=1}^M N_m^1 : \sum_{m=1}^M N_m^2 : \cdots : \sum_{m=1}^M N_m^k$ to denote the global class imbalance. These two measures might be dramatically diverse as random participants generate two different proportions, called local, and global class imbalance inconsistency.

Quantifying the degree of non-IID is still an open problem with no uniform standard. The magnitude of the difference between the minimum global objective function and the minimum weighted average of the local objective function is provided to measure this heterogeneity [17]. This is a very general method that includes situations of all types of non-IID data and all object function performance in Fedavg. However, it cannot guide us on how to select participants' data distribution and analyze the impact of heterogeneity degree on training results. Thus, we only utilize the L2 norm to describe the class distribution in every participant that we regard τ_m as a vector normalized from ρ_m

$$\nabla\Gamma = \left| \Gamma_{\text{global}} - \sum_{m=1}^{M} a_m \Gamma_m \right|, \quad \Gamma_m = \frac{\|\tau_m\|_2 \cdot \sqrt{K} - 1}{\sqrt{K} - 1} \tag{5}$$

where $\tau_m^k = N_m^k / \sum_{k=1}^K N_m^k$ is the factor of class k proportion in τ_m. Γ_{global} is the imbalance criteria corresponding to the global percentage ρ_{global} based on the same algorithm with Γ_m. The lower limit of Γ_m is class label uniform distribution, and the upper limit corresponds to the one-hot vector representing every participant having one class. The lower Γ_m scenario is more likely to belong to horizontal FL; the higher Γ_m is more like federated transfer learning. Comparison between the local and global imbalance in FL can be adjusted according to different aggregation functions, such as Fedavg, FedProx [41], and FedNova [21], which will have various a_m. If data are IID, $\nabla\Gamma$ obviously reaches to zero. If data are non-IID, $\nabla\Gamma$ represents the degree of global and local imbalance inconsistency adjusted by the aggregation function.

IV. Proposed Synergetic Focal Loss

The graphical interpretation of our method is depicted in Fig. 1. We replace the conventional cross-entropy with the focal loss for class imbalance objective function adjustment in multiclassification with federated XGBoost. Sequentially, we propose a hyperparameter determinate strategy, synergetic focal loss, and illustrate its impact on global and local class imbalance.

A. *Focal Loss for Multiclassification*

The binary classification generally uses a sigmoid activation function in the cross-entropy loss. Any multiclassification task can be transformed into several one-versus-all binary classifications to predict target probability over the rest of the classes. Transformed multiclassification is appropriate for independent and not mutually exclusive multidimensional classes, which might have several semantic labels [40]. For example, a tomato can be a fruit or a vegetable. Unlike the sigmoid function, softmax solves the mutual exclusion of multiple classifications that cooperate with one-hot real label transformation, such as handwritten digit recognition and premium rank classification. When applied to the multiclassification problem, the difference of softmax classes probability is relatively significant, and the maximum probability value close to 1 means the model performed well. Hence, the form of the output distribution is closer to a one-hot matrix. When the class is 2, it degenerates into binomial distribution. Softmax uses exponential functions to amplify the differences among the input vector elements and then normalize them into a probability distribution. The softmax cross-entropy loss can be written as

$$l\ \left(y_m^i, p_m^i\right) = -\sum_{k=1}^{K} y_m^{ik}\log\left(p_m^{ik}\right),\ \ p_m^{ik} = \frac{e^{\hat{y}_m^{ik}}}{\sum_{k=1}^{K} e^{\hat{y}_m^{ik}}} \quad (6)$$

where y_m^i is the transformation of the one-hot matrix from multiple classes vector, p_m^i is the probability matrix corresponding to each class. Sequentially, obtaining the derivation $\partial p_m^{ik}/\partial \hat{y}_m^{ih} = p_m^{ih}\left(1 - p_m^{ih}\right)$ (if $k = h$); $\partial p_m^{ik}/\partial\ \hat{y}_m^{ih} = -p_m^{ik} p_m^{ih}$ $(if\ k! = h)$. Finally, total loss function derivation is

$$\text{Gradient}:$$

$$\frac{\partial l}{\partial \hat{y}_m^{ih}} = -\sum_{k=1}^{K} y_m^{ik} \frac{1}{p_m^{ik}} \frac{\partial p_m^{ik}}{\partial \hat{y}_m^{ih}} = \begin{cases} p_m^{ih} - 1\ (\text{if}\ \ k = h) \\ 0\ \ (\text{if}\ k! = h) \end{cases} \quad (7)$$

$$\text{Hessian}:\ \frac{\partial^2 l}{\partial (\hat{y}_m^{ih})^2} = \begin{cases} p_m^{ih}\left(1 - p_m^{ih}\right)\ (\text{if}\ \ k = h) \\ 0\ \ (\text{if}\ k! = h) \end{cases}. \quad (8)$$

Training models prefer to predict all samples as majority classes to gain high accuracy in extreme imbalance scenario; ignoring minority classes become straightforward. Thus, samples with the wrong predictions should be given more weight in subsequent learning with the advantages of XGBoost, an integration of a series of weak classifiers. The focal loss was initially utilized to solve the class imbalance in object detection as the number of targets is small [42]. It is a dynamically scaled cross-entropy loss applied to two modulating terms instead of the basic cross-entropy loss. $\left(1 - p_m^{ik(t)}\right)^{\gamma}$ is used to decay scaled factors dramatically of correct classification samples and keep them predicting constantly correct on the following training, focusing the model on the hard minority samples through slightly down-weight the misclassification simultaneously. $\alpha_k^{(t)}$ adjusts the loss after $\left(1 - p_m^{ik(t)}\right)^{\gamma}$ coefficient attenuation. Focal loss in multiclass imbalanced problem can be expressed as

$$FL_{\text{softmax}}^{(t+1)} = -\sum_{k=1}^{K} \alpha_k^{(t)} \left(1 - p_m^{ik(t)}\right)^{\gamma} y_m^{ik} \log\left(p_m^{ik(t)}\right)$$
$$p_m^{ik} = \frac{e^{\hat{y}_m^{ik}}}{\sum_{k=1}^{K} e^{\hat{y}_m^{ik}}}. \quad (9)$$

In (9), γ is a new hyperparameter to control the strength of managing the shape of the loss function curve. When $\gamma = 0$, this loss is equivalent to weighted cross-entropy loss. α originally is the inverse ratio of sample proportions assigned to the minority class. However, with a powerful change from exponent γ, lower α should be matched to higher γ and vice-versa. As we have seen, γ plays the dominant role in focal loss.

Applying focal loss to federated XGBoost, the appropriate gradient and hessian should be derived. The hessian expression is too complex to be omitted here (see Appendix)

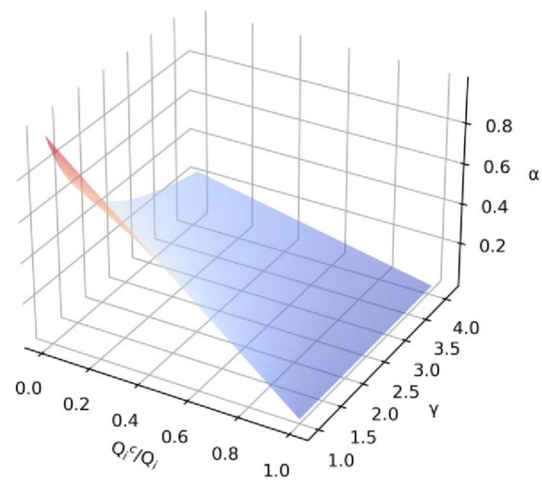

Fig. 2. Relationship among class percentage and two hyperparameters of (11).

$$\text{Gradient}: \frac{\partial \text{FL}}{\partial \hat{y}_m^{ih}} =$$
$$\begin{cases} D \cdot \left[\gamma p_m^{ih(t)} \log\left(p_m^{ih(t)}\right) - 1 + p_m^{ih(t)}\right]\ (\text{if}\ k = h) \\ E \cdot \left[p_m^{ih(t)} - p_m^{ik(t)} p_m^{ih(t)} \left(\gamma \log\left(p_m^{ik(t)}\right) - 1\right)\right] (\text{if}\ k! = h) \end{cases} \quad (10)$$

where $D = \alpha_k^{(t)} y_m^{ik} (1 - p_m^{ih(t)})^{\gamma}$, $E = \alpha_k^{(t)}\ y_m^{ik} (1 - p_m^{ik(t)})^{\gamma - 1}$.

We will find the splitting feature and value of the next tree node through the above operation according to the gain criteria in XGBoost. Then, the leaf score of the new tree will be recalculated, and the probability corresponding to each sample will be obtained according to the Softmax function.

B. Synergetic Focal Loss for Hyperparameters Determination

In terms of the multiclassification problem, optimizing focal loss with an appropriate α_k for each class is almost impossible by trying different numbers and gusting complex relationships in all classes. We find α_k has a strong correlation with γ and the number proportion of class k in P_m. Then, we proposed a computable algorithm for α_m^k of each P_m to modify the hyperparameter from focal loss

$$\alpha_m^k = \frac{1 - N_m^k / N_m}{\gamma} \quad (11)$$

where γ is selected by the global server and transmitted to all participating clients in a certain iteration. In FL, N_m^k is the number of class $k \in K$ in P_m, N_m^k / N_m is the proportion of class k in P_m based on quantity weight. Through this formula, α inversely related to the class frequency and γ. Synergetic focal loss does not change (9) property that contains cross entropy, weighted focal loss, and other variants. We can see that when the class proportion is low and γ is close to 1, α is close to 1 in Fig. 2, which is transformed as a cross-entropy loss function. From this

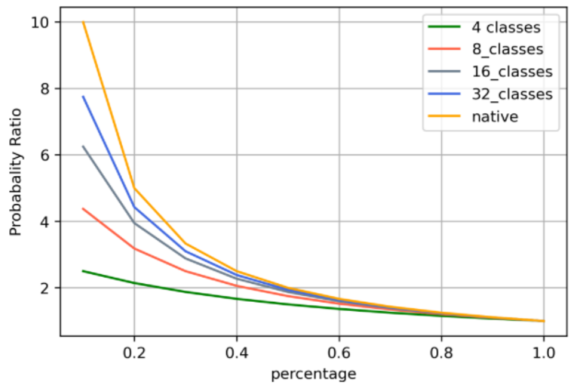

Fig. 3. Ratio of majority and minority classes present ratio with LLT partition for different global class numbers.

perspective, the selection of hyperparameters α, γ can be slid from the blue part representing synergetic focal loss searching to the red part corresponding to native federated XGBoost.

In this way, each participant can obtain specific α_m^k for all classes based on local data distribution without transmitting any statistics to the server or other participants. Also, the server passes γ without additional communication to learn the tedious situations such as inconsistency between global and local imbalance. This setup fits well within the FL scenario. The ratio adjustment after normalized weight w_m^k between two different classes is

$$w_m^k = \frac{\alpha_m^k}{\sum_{k\epsilon P_m}\alpha_m^k}\ ,\quad \frac{w_m^{k_1}}{w_m^{k_2}} = \frac{\alpha_m^{k_1}}{\alpha_m^{k_2}} = \frac{N_m - N_m^{k_1}}{N_m - N_m^{k_2}}\ . \tag{12}$$

Clients only calculate the new sample weights locally before training starts, ensuring the privacy of FL's original design objective. The weight after normalization landed on $[0, 1)$, combined with the characteristics of data volume, facilitate the weighted average aggregation of the different numbers of training clients who participated in each round [21]. Assume k_1 and k_2 are one of minority and majority class, class sampling, the probability of class k that can be selected in client m, is $\delta_m^k = N_m^k \cdot w_m^k$. The ratio of any two classes presented in P_m is adjusted as

$$\frac{\delta_m^{k_1}}{\delta_m^{k_2}} = \frac{N_m^{k_1}\cdot w_m^{k_1}}{N_m^{k_2}\cdot w_m^{k_2}} = \frac{N_m^{k_1}\cdot\left(N_m - N_m^{k_1}\right)}{N_m^{k_2}\cdot\left(N_m - N_m^{k_2}\right)}. \tag{13}$$

Assuming we only have just two classes with 100 and 900 samples, the weight ratio becomes $9:1$. The probability ratio of two classes in (13) compressed the curvature of the inverse classes' frequency method. So, it alleviates the potent effects of γ. The sample is readjusted according to the quantity proportion when it comes to the multiclassification. Moreover, the more classes that should be judged, the closer to the original ratio. This indicates that the greater the difference in quantity between classes, the more adjustment is needed; The more kinds of classes to be judged, the more difficult the judgment will be, and the corresponding adjustment will be increased.

Adjustment impact based on local long-tail (LLT) partition [32] shows in Fig. 3. Furthermore, aggregated class sampling from the global perspective has also become more similar. In this way, the local imbalance problem that also happens in traditional ML scenarios got adjusted effectively. The unique problem of local and global class imbalance inconsistency in FL has also been mitigated. Moreover, each participant does not need to find a standard alignment that alleviates the global imbalance. The caveat here is that this approach does not work well if many participants have massive missing classes, which more likely belong to federated transfer learning.

Algorithm 1: Synergetic Focal Loss in Federated XGBoost Learning Approach.

Input: Pre-processed Client Dataset
Output: Multi-classification Model
$************$Procedure Server-Update $*************$
Initialize the synergetic focal loss function $l_{softmax}^{fl}$ with certain start γ^*, interval φ and XGBoost algorithm
Tree $\leftarrow$ { }
Criteria $\leftarrow 0$
For $\gamma = \gamma^*$ do
  Client-Updated (g_i^j, h_i^j) :
  Training XGBoost Classifier on t' trees;
  Receive criteria results (E.g accuracy, error)
  If criteria results $>$ Criteria (assume the higher; the better)
    Criteria $\leftarrow$ criteria results
    $\gamma \leftarrow \gamma - \varphi$;
    continue
  Else
    break
End
Return the best accuracy corresponding γ;
Tree $\leftarrow$ append t' trees
$*************$Procedure Client-Update $************$
For each client do
  Receive γ and calculate α_i^{jc} for each class
  Calculate g_i^j and h_i^j based on Splitting Matrix
Return corresponding g_i^j, h_i^j
Receive the best performance decision trees
Calculate local training accuracy, F1-score, Recall …

In terms of the computational complexity of synergetic focal loss utilized in federated XGBoost, training time complexity has been increased $O(cMk)$ from the original synchronous federated XGBoost's $O(cdxlogN)$ and $O(Mk)$ from focal loss with federated XGBoost since it should compute every participant's local α_m vector, where c is the count of trees, d is the height we set for every tree, x is the number of total training data without missing entries. In the new algorithm, the computational complexity of the parameter α increases with the number of participants and the number of categories to be classified. Training space complexity is similar to with or without focal loss in federated XGBoost.

The relationship between the two kinds of hyperparameters is clearly described, but the suitable γ is a difficult one to decide. We know γ is within a certain range from previous experiments [34], [42], and it does not behave well with too large or too small value, which will be confirmed by our experimental results in Section V. Thus, we can go through a loop to find the γ and corresponding α_m^k that are best suited for a particular problem in the establishment of a finite number of trees. γ^* in Fig. 1 is the choice when the global accuracy is at its highest, whereas it might not be the fact for each participant. Different α_m^k for same class in different participants denote slightly different weights. In FL, a fixed node always defaults as the server, or one of the participants can act as a proxy server since it is the initiator of the task. We always use the proxy server to represent. In the second set, that participant performs local training as other participants do in addition to server aggregation tasks. Algorithm 1 is the supplementary instruction for the flowchart that illustrates the detailed transmission and calculation process.

V. Proposed Synergetic Focal Loss Simulation and Results

A. Simulation Scenario

The risk classifies identification of customers' eligibility always takes about one month before getting a reasonable quote, which is too long, labor-consuming, and turns people OFF. The one-click shopping demand market requires a significantly streamlined insurance process for a more accurate automatic risk classification without sacrificing customers' privacy. This scenario is a perfect match for applying federated XGBoost with multiclassification focal loss.

B. Dataset and Preprocessing

A multiclassification experiment is carried out by Prudential's life insurance risk assessment dataset [43], which is one of the largest issuers of life insurance in the USA. The dataset has 59 381 samples (28 observations repeated) and 126 describing attributes, such as basic health information, employment history, insurance history, and medical history that are intensely private for personalization. They can be separated into 13 continuous variables, 5 discrete variables, 60 categorical variables, and 48 dummy variables. Different kinds of variables should be treated with corresponding manners of preprocessing. We utilize Box-Cox to change continuous variables to normalized distribution since some are skewed. Discrete variables are treated in the same manner as continuous variables. Dummy variables do not need any preprocessing. There are no missing values for categorical variables, and we transformed them into dummy variables. However, this operation makes the whole variable matrix extremely sparse. It is challenging to extract useful information for machine learning models. Hence, we apply principal component analysis (PCA) to the processed categorical variables and dummy variables, which can not only save the computational cost of participants but also prevent malicious attackers from inferring original data information through training gradient. With this simulation, we explored a federated learning environment without good hardware and with reduced fidelity requirements. All missing values have been retained intact because XGBoost itself will confirm whether the left or right branch of the missing value sample belongs according to the gain expressed in (4). Responses are an ordinal measurement of eight-level risks, but they are not rank targets. The task of our experiments is to predict this variable of each customer.

C. Experiments Results

We conduct the simplest experiments based on FL XGBoost created by the UC Berkeley RISE Lab [44] with two parties; each can initiate tasks as the server. Typically, horizontal FL in the same domain occurs among large institutions, such as hospitals and banks. In this case, the number of participants is very limited. In addition to FL in a typical star topology, decentralized FL aggregations have designed ring and mesh topologies. They have been thoroughly analyzed in terms of the rate of communication to computational complexity, accuracy, and heterogeneous data distribution application [45], [46]. SimFL also shared GBDT built from one party to others sequentially [8]. All these architectures can be reduced to a point-to-point exchange of parameters between two participants. Fig. 4 shows the visualization of three different ways of data partition as each party selects 12 000 samples. The first row of this partition indicates eight-level risks, which are also our classification targets. The first column means aggregated and local participants. The data distribution ratios to all and each participant are displayed in it. Fig. 4(a) is a random selection for all participants that class proportions are similar to global distribution. According to the quantification, we proposed in (5), $\Gamma_{\text{global}} = 0.136359$, $\nabla\Gamma = 0.000157$. Fig. 4(b) determined by us that maximizes the usage of minority classes in the raw dataset and all classes have samples in each party to create a coexisting global and local class imbalance scenario, which is not extremely imbalanced. Calculated results are $\Gamma_{\text{global}} = 0.085301$, $\nabla\Gamma = 0.042886$. We get that the distribution is more uniform globally, but the participants' and the global's distributions are significantly more different. This outcome cannot be controlled by us as the amount of raw data is limited. Then, we create a new scenario Fig. 4(c), determined by us as Fig. 4(b), but set some missing classes in each participant. $\Gamma_{\text{global}} = 0.075921$ and $\nabla\Gamma = 0.049041$ means more even global distribution and difference among participants and aggregated global data. These two degrees are relatively insignificant between two self-determined datasets. This partitioning method better fits the randomness of participants' data in the real FL environment and fully applies the existing real unbalanced life insurance data.

In terms of PCA, we set 10%, 30%, and 50% of variance that need to be explained. With randomly selected dataset distribution, the experiment results of one participant are shown in Table I. Federated XGBoost with focal loss (Fed-FL-XGBoost) sets $\gamma = 2$, $\alpha_m^k = 0.25$. Our proposed federated XGBoost with synergetic focal loss (Fed-SynFL-XGBoost) gives different α_m^k also based on $\gamma = 2$. These settings are approximately equal to the consistency between global and local imbalances. We can see that our method considerably accelerates the ultimate criteria. The bold numbers indicate the best performance of each criterion among the different methods. Other tables are similar. Since we estimate multiclassification, the weighted average of precision, recall, and F1-score are more recommended. Although the distribution of each class is uneven, the high precision, recall, and F1 score indicate that minority classes are not ignored in this model.

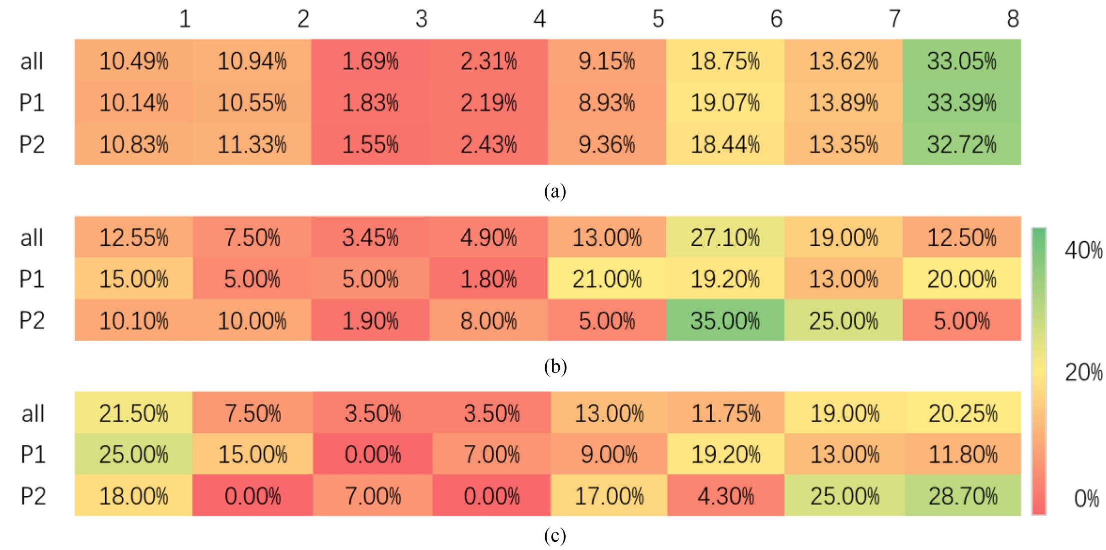

Fig. 4. Original dataset distribution and two different data partition distributions.

TABLE I
WEIGHTED AVERAGE PERFORMANCE FOR DIFFERENCE PCA. SETTING WITH RANDOM SAMPLING

PCA.	Method	Precision(WA.)	Recall(WA.)	F1-score(WA.)	Accuracy
10%	Native Fed-XGBoost	0.7073	0.6913	0.6857	0.6913
	Fed-FL-XGBoost	0.766	0.7583	0.7541	0.7583
	Fed-SynFL-XGBoost	**0.7837**	**0.7813**	**0.7787**	**0.7813**
30%	Native Fed-XGBoost	0.7819	0.7687	0.7673	0.7687
	Fed-FL-XGBoost	0.8016	0.7943	0.7919	0.7943
	Fed-SynFL-XGBoost	**0.8325**	**0.8297**	**0.8288**	**0.8297**
50%	Native Fed-XGBoost	0.7993	0.7874	0.7865	0.7874
	Fed-FL-XGBoost	0.8109	0.8043	0.8024	0.8043
	Fed-SynFL-XGBoost	**0.842**	**0.8396**	**0.8389**	**0.8396**

The bold numbers indicate the best performance.

At the same time, the higher the PCA percentage, the better improvement of the four criteria and the model's performance, which indicates that our method can well absorb the information in the data and PCA does work when participants have saving storage consumption and special privacy protection needs.

The performance of the proposed method is shown in Table II with self-determined dataset distribution, including with or without class absent scenarios. In the case of global and local imbalance inconsistency, all settings are similar with randomly selected datasets. Synergetic focal loss for Fed XGBoost still totally outperformed basic methods on four criteria. Fed XGBoost without class absence is obviously better than that with class absence in the condition that global and inconsistency imbalance quantification are close. Additionally, our approach's improvement of low variance PCA is more remarkable without class absence. The class absence will partially offset this functionality. Comparing Tables I and II, we will find that all methods in the self-determined dataset without class absence still perform better even weighted local class imbalance quantifications are similar for three partitioning distributions. There are two possible reasons: the g global class imbalance in the self-determined sampling dataset drops significantly, and the participants without missing class do not need to learn external information through the common model issued in the last round. Such a global model already dilutes the unique class learned by a participant in an aggregation. Moreover, the self-determined dataset with class absence performs slightly worse than the randomly selected dataset, indicating that the absence of different class examples in various participants does pose some problems for federated aggregation. However, the improvement of the global sample balance does not offset the impact of class absence.

Convergence of Three Methods. All the solid lines in Fig. 5 are from the randomly selected dataset; all the big, dotted lines are from self-determined datasets without class absence; and all the small, dotted lines are from self-determined datasets with class absence. According to the curves, native federated XGBoost always starts from a high prediction error. Fecal loss and synergetic focal loss for FL XGBoost hold a relatively low prediction error rate from the beginning. Moreover, the convergence rate increases successively from native federated XGBoost to the synergetic focal loss for FL XGBoost. Fecal loss and synergetic focal loss for FL XGBoost reach the stable

TABLE II
WEIGHTED AVERAGE PERFORMANCE FOR DIFFERENCE PCA. SETTING WITH SELF-DETERMINED SAMPLING

	PCA.	Method	Precision(WA.)	Recall(WA.)	F1-score(WA.)	Accuracy
Without Class Absent	10%	Native Fed-XGBoost	0.7164	0.6798	0.6856	0.6798
		Fed-FL-XGBoost	0.7672	0.7484	0.7448	0.7484
		Fed-SynFL-XGBoost	**0.8072**	**0.7938**	**0.7937**	**0.7938**
	30%	Native Fed-XGBoost	0.7902	0.7714	0.7749	0.7714
		Fed-FL-XGBoost	0.8062	0.7944	0.7927	0.7944
		Fed-SynFL-XGBoost	**0.8493**	**0.8415**	**0.8419**	**0.8415**
	50%	Native Fed-XGBoost	0.8027	0.7872	0.7902	0.7872
		Fed-FL-XGBoost	0.8132	0.8022	0.8006	0.8022
		Fed-SynFL-XGBoost	**0.8572**	**0.8504**	**0.8511**	**0.8504**
	PCA.	Method	Precision(WA.)	Recall(WA.)	F1-score(WA.)	Accuracy
With Class Absent	10%	Native Fed-XGBoost	0.6913	0.6221	0.6266	0.6221
		Fed-FL-XGBoost	0.737	0.6949	0.6857	0.6949
		Fed-SynFL-XGBoost	**0.7667**	**0.7320**	**0.7330**	**0.7320**
	30%	Native Fed-XGBoost	0.7715	0.7301	0.736	0.7301
		Fed-FL-XGBoost	0.7861	0.7563	0.7527	0.7563
		Fed-SynFL-XGBoost	**0.8239**	**0.8030**	**0.8059**	**0.8030**
	50%	Native Fed-XGBoost	0.8149	0.7864	0.7919	0.7864
		Fed-FL-XGBoost	0.7942	0.7658	0.7635	0.7658
		Fed-SynFL-XGBoost	**0.8293**	**0.8087**	**0.8115**	**0.8087**

The bold numbers indicate the best performance.

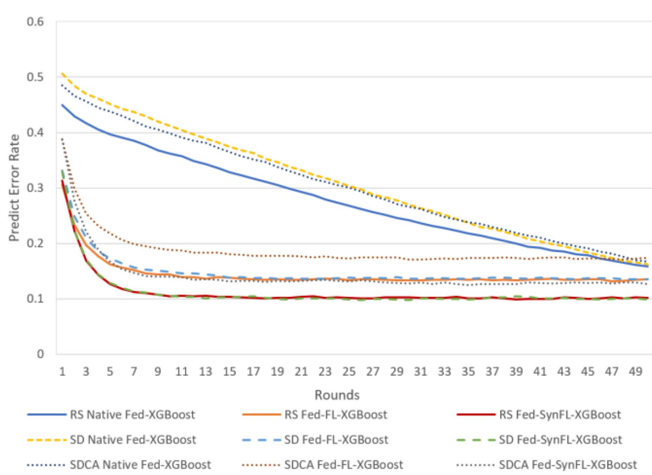

Fig. 5. Comparison of predicted error percentage during XGBoost tree building rounds with 30% PCA.

prediction tree model on the 10th round, significantly reducing the computation and communication required for model updates. For Native Fed-XGBoost, self-determined partition with or without class absence lags behind the accuracy of random sampling on the whole until 50 communication rounds. However, for Fed-FL-XGBoost and Fed-SynFL-XGBoost, the accuracy of self-determined sampling without class absence is almost equivalent to that of random sampling, It indicates that synergetic focal loss inherits the correction of focal loss for imbalanced class distribution. Simultaneously, Fed-SynFL-XGBoost improves the accuracy on the whole.

Effect of α_m^k. All experiment results in Table III are based on the 30% PCA variance. Different γ corresponds to specific α_m^k in participant P_m considering class number proportion. All criteria are stable without sharp ups and downs, indicating an effective overall performance of synergetic focal loss. Although the global γ corresponding optimal results of the three data partitions differ, they are all working well as examples of the trade-OFF of γ and α. Then, we can run a for loop to find the best γ for all participants according to Algorithm 1. The outcome shows that different participants should adopt matched γ for their own if they want to pursue extreme precision, which might be no need in the real world. On the one hand, the calculation amount brought by for loop is several times or even tens of times of the original selecting several points, which does not meet the requirements of FL to minimize information transmission. On the other hand, the performance of for loop is not significantly better than the selected points in the experimental results, it is not worth to take a lot of effort. So proposed method provides a reference for multi-classification hyperparameter selection in the focal loss.

D. Experimental Extension

Three more datasets are used to conduct three additional experiments to calculate quantifications of the class imbalance inconsistency, evaluate the effectiveness of synergetic focal loss and the improvement of the existing federated XGBoost. The

TABLE III
DIFFERENT γ IN RANDOM SAMPLING AND SELF-DETERMINED DATASETS

	γ	Precision(WA.)	Recall(WA.)	F1-score(WA.)	Accuracy
Ran-Sam	1.5	0.8077	0.8042	0.8026	0.8042
	2.0	0.8325	0.8297	0.8288	0.8297
	2.5	**0.8339**	**0.8313**	**0.8303**	**0.8313**
	3.0	0.8238	0.8206	0.8193	0.8206
Best	2.3	0.8379	0.8348	0.8342	0.8348
	2.4	0.8369	0.8345	0.8336	0.8345
Self-Det Without Class Absent	1.5	0.8248	0.8112	0.8128	0.8112
	2.0	**0.8493**	**0.8415**	**0.8419**	**0.8415**
	2.5	0.8464	0.8391	0.8389	0.8391
	3.0	0.8347	0.8251	0.8246	0.8251
Best	2.3	0.8379	0.8348	0.8342	0.8348
	2.2	0.8489	0.8424	0.8426	0.8424
Self-Det With Class Absent	1.5	0.797	0.7649	0.7693	0.7649
	2.0	**0.8239**	**0.803**	**0.8059**	**0.803**
	2.5	0.8228	0.8018	0.8041	0.8018
	3.0	0.8057	0.7798	0.7804	0.7798
Best	2.4	0.8236	0.8038	0.8063	0.8038
	2.1	0.8615	0.8544	0.8569	0.8544

The bold numbers indicate the best performance.

first dataset is tabular binary classification; the other two are image-based multiclassifications. In the following experiments, we use randomly selected subsets of datasets either due to slow baselines or to demonstrate the method's performance for higher quantified class imbalance. See Fig. 6, 7 and 8 in the appendix for specific data distributions.

HIGGS Boson: The data is produced by Monte Carlo simulation, distinguishing, which events produced Higgs bosons. 21 features are kinematic properties measured by the particle detectors in the accelerator, and the other 7 are high-level features derived from physicists. We randomly select 10 M instances to federate amongst the participants. This task is a binary classification.

MNIST: The data includes 0 to 9 handwritten digits in fixed-size grayscale images containing 60 000 training sets and 10 000 test sets. As we will federate the dataset among participants to simulate the label imbalance scenario in the real world, not all the balanced digital records will be used in our experiments.

Chinese MNIST: In total, 15 classes of Chinese characters representing digital measurement units scanned by 64×64 size image formed a 15 000 experimental dataset. The handwriting in images is very fine, and there is much invalid information on the edge. Thus, we selected [20:57, 17:47] as target classification information. Similar to MNIST, we cannot use a fully balanced dataset. Characters' associated values can be transformed into 0 to 14 class labels.

All experiments are conducted on VMware virtual machines in Nitro AN515-45 with AMD Ryzen 9 5900HX Radeon Graphics 3.30 GHz and 32 G memory. The maximum depth of all boost trees is settled as 6, and the learning rate is 0.08. Subsample and column subsampling are 0.8 unless explicitly specified. For datasets, we simulate four scenarios: native random selection for all participants (RS), self-determined class distribution without label absent (SD), and with zero number of several classes (SDCA) like the partition mechanism in Fig. 4 The last one is LLT mentioned in Fig. 3 that setting the ratio of major and minor classes, and with only one major class, the rest are the same number of minor class in each participant. Here, we set it as 8:2. Training sets maximum use abovementioned public datasets and the remaining data is subsumed into testing data. Since manufacturing the missing class for binary classification in HIGGS, the training scenario would convert to federated transfer learning. If utilizing our proposed method, we should cancel it because the training experiment will fail to converge. The class label distribution details in three datasets are attached in Appendix. The corresponding degree of global and local imbalance quantification is revealed in Table IV.

The original distributions of the three datasets are uniform. Self-determined sampling increases the local imbalance, but the global class imbalance decreases with a combination. In addition, self-determined sampling with class absence expands the inconsistency of global and local class imbalance. For LLT, the more global classes there are, the more balanced its data distribution will be since more minority classes will dilute the proportion of only one majority class.

Table V is the collection of all three partition methods with three datasets' experimental results. All hyperparameters in XGBoost are selected from the centralized one in Kaggle based on the same datasets. From the results, our methods are generally better than benchmark models: Random Forest and GBDT in FL,

TABLE IV
DEGREE OF GLOBAL AND LOCAL IMBALANCE FOR THREE SCENARIOS

	HIGGS Boson			MNIST				Chinese MNIST			
Scenario	RS	SD	LLT	RS	SD	SDCA	LLT	RS	SD	SDCA	LLT
Γ_{global}	0.003729	0.000000	0.000000	0.000000	0.058145	0.074829	0.046882	0.000000	0.007248	0.018635	0.030122
$\sum_{m=1}^{M} p_m \Gamma_m$	0.003771	0.487757	0.401219	0.000000	0.093354	0.143200	0.100015	0.000174	0.040230	0.066005	0.062136
$\nabla \Gamma$	0.000042	0.487757	0.401219	0.000000	0.035209	0.068371	0.053133	0.000174	0.032982	0.047371	0.032014

The bold numbers indicate the best performance.

TABLE V
PERFORMANCES OF THREE DATASETS IN THREE SCENARIOS

		HIGGS Boson			MNIST				Chinese MNIST			
Scenario		RS	SD	LLT	RS	SD	SDCA	LLT	RS	SD	SDCA	LLT
Precision(WA)	Random Forest	0.6538	0.6818	0.6634	0.7636	0.5880	0.6940	0.7643	0.6515	0.7002	0.5930	0.6627
	GBDT	0.6997	0.7219	0.7077	0.8717	0.6924	0.7422	0.8584	0.6917	0.6973	0.5917	0.6871
	Native Fed-XGBoost	0.7909	0.7472	0.7436	0.9664	**0.9006**	0.8627	0.7514	0.7762	0.7446	0.7396	0.7568
	Fed-FL-XGBoost	0.8188	0.7692	0.7119	0.9317	0.8044	0.8562	0.8341	0.7800	0.7613	0.7939	0.6565
	Fed-SynFL-XGBoost	**0.8364**	**0.8003**	**0.7535**	**0.9712**	0.8124	**0.8817**	**0.8717**	**0.8195**	**0.7754**	**0.8259**	**0.7614**
Recall(WA)	Random Forest	0.6541	0.6464	0.6494	0.7440	0.5574	0.6693	0.7369	0.6402	0.5363	0.5169	0.5244
	GBDT	0.7001	0.6961	0.6994	0.8551	0.6123	0.7051	0.8533	0.7010	0.5912	0.5534	0.5599
	Native Fed-XGBoost	0.7911	0.7302	0.7380	0.9662	0.7579	0.7760	0.7305	0.7753	0.6809	0.6278	0.6658
	Fed-FL-XGBoost	0.8190	0.7534	0.7076	0.9294	0.7258	0.8521	0.7753	0.7807	0.7208	0.6758	0.6533
	Fed-SynFL-XGBoost	**0.8260**	**0.7839**	**0.7476**	**0.9704**	**0.7873**	**0.8803**	**0.8551**	**0.8287**	**0.7798**	**0.7350**	**0.7539**
F1-score(WA)	Random Forest	0.6539	0.6534	0.6513	0.7407	0.5510	0.6239	0.7368	0.6178	0.5055	0.5022	0.5351
	GBDT	0.6997	0.7018	0.7010	0.8557	0.6169	0.7804	0.8541	0.6874	0.5522	0.5250	0.5883
	Native Fed-XGBoost	0.7910	0.7346	0.7393	0.9662	0.7615	0.7666	0.7296	0.7743	0.6961	0.6527	0.7015
	Fed-FL-XGBoost	0.8188	0.7577	0.7088	0.9298	0.7242	0.8533	0.7468	0.7789	0.7208	0.6937	0.6190
	Fed-SynFL-XGBoost	**0.8261**	**0.7884**	**0.7491**	**0.9705**	**0.7831**	**0.8805**	**0.8557**	**0.8135**	**0.7849**	**0.7399**	**0.7357**
Accuracy	Random Forest	0.6541	0.6464	0.6494	0.7440	0.5574	0.6693	0.7369	0.6402	0.5363	0.5169	0.5244
	GBDT	0.7001	0.6961	0.6994	0.8551	0.6123	0.7051	0.8533	0.7010	0.5912	0.5534	0.5599
	Native Fed-XGBoost	0.7911	0.7302	0.7380	0.9662	0.7579	0.7760	0.7305	0.7753	0.6809	0.6278	0.6658
	Fed-FL-XGBoost	0.8190	0.7534	0.7076	0.9294	0.7258	0.8521	0.7753	0.7807	0.7218	0.6758	0.6533
	Fed-SynFL-XGBoost	**0.8260**	**0.7839**	**0.7476**	**0.9704**	**0.7873**	**0.8803**	**0.8551**	**0.8287**	**0.7798**	**0.7350**	**0.7539**

Native Fed-XGBoost with cross-entropy, and Fed-FL-XGBoost to varying degrees. However, focal loss replacement has no advantage when the cross-entropy has performed well for balanced data, such as MNIST with SD partition and Chinese MNIST with LLT partition. GBDT sometimes also performs well when MNIST with LLT partition. Our method can only undertake a remediation role and does not significantly improve it. Simultaneously, the effect of focal loss and synergetic focal loss on binary classification is insignificant. Experimental results also verify that our approach is more matching for multiclassification datasets. For self-determined class imbalance data, FL has limited influence on tabular data but enormous repercussions on image classification. It may be that the matrix information in image data is sparse, so the amount of data becomes crucial for XGBoost decision tree classification. In terms of Chinese MNIST, testing performs worse than CNN and other deep learning models in SD, SDCA, and LLT scenarios due to insufficient data.

For the significance of experimental results, Spearman's Rank Correlation and Chi-square test for Accuracy and F1-score were used to detect the correlation between Fed-FL-XGBoost and Fed-SynFL-XGBoost. Then, after the Levene test was used to test the homogeneity of variances between the two methods, p values were obtained as 3.9431×10^{-9} and 6.4648×10^{-8} respectively by paired sample t-test. These values indicate significant differences between the means of the two methods. Both methods' experimental results show highly correlated, which was in line with what we expected as synergetic focal loss is a modified version of focal loss used in federated XGBoost. Meanwhile, the newly proposed synergetic focal loss is significantly superior to focal loss applied to federated XGBoost.

VI. CONCLUSION AND FUTURE WORK

In this article, we deeply explore the application of multiclassification in federated XGBoost and introduce focal loss as a scheme to improve class imbalance. Based on the complex global and local imbalance problem in FL, we propose a delicate method to determine hyperparameters in focal loss, making the model convergence speed significantly faster; Precision, Recall, F1-score, and accuracy all perform remarkably. Simultaneously, this method meets the FL privacy protection requirements and does not add extra data communication and computation. We

$$\text{Gradient}: \frac{\partial \mathrm{FL}}{\partial \hat{y}_m^{ih}} = \begin{cases} \alpha_k^{(t)} y_m^{ik} \left(1 - p_m^{ih(t)}\right)^{\gamma} \left[\gamma p_m^{ih(t)} \log\left(p_m^{ih(t)}\right) - 1 + p_m^{ih(t)}\right] & (\text{if } k = h) \\ \alpha_k^{(t)} y_m^{ik} \left(1 - p_m^{ik(t)}\right)^{\gamma - 1} \left[p_m^{ih(t)} - p_m^{ik(t)} p_m^{ih(t)} \left(\gamma \log\left(p_m^{ik(t)}\right) - 1\right)\right] & (\text{if } k != h) \end{cases} \tag{14}$$

$$\text{Hessian}: \frac{\partial^2 \mathrm{FL}}{\partial \left(\hat{y}_m^{ih}\right)^2} = \begin{cases} \alpha_k^{(t)} y_m^{ik} p_m^{ih(t)} \left(1 - p_m^{ih(t)}\right)^{\gamma} \left[\gamma \left(1 - p_m^{ih(t)} - \gamma p_m^{ih(t)}\right) \log\left(p_m^{ih(t)}\right) - p_m^{ih(t)} + 2\gamma + 1\right] & (\text{if } k = h) \\ \alpha_k^{(t)} y_m^{ik} p_m^{ih(t)} \left(1 - p_m^{ih(t)}\right)^{\gamma - 2} \begin{bmatrix} \gamma p_m^{ik(t)} \log\left(p_m^{ik(t)}\right) \left(p_m^{ik(t)} p_m^{ih(t)} (\gamma - 1) + p_m^{ik(t)} - 1\right) + \\ p_m^{ik(t)} p_m^{ih(t)} - {p_m^{ik(t)}}^2 p_m^{ih(t)} + {p_m^{ih(t)}}^2 - {p_m^{ik(t)}}^2 - p_m^{ih(t)} + 1 \end{bmatrix} & (\text{if } k != h) \end{cases} . \tag{15}$$

conduct extensive experiments to demonstrate proposed method reaches a high-level stable to give us a reference.

We did not give a reasonable solution for binary classification in the extreme imbalance scenario where one class only appeared in a particular participant to make our method more broadly applicable. Extending global and local imbalance solutions to a more general ML also leads researchers to study further.

Appendix

	0	1
all	47.22%	52.78%
P1	46.97%	53.03%
P2	47.46%	52.54%

(a)

	0	1
all	49.99%	50.01%
P1	83.33%	16.67%
P2	16.63%	83.37%

(b)

Fig. 6. Classes distribution of HIGGS for two participants. (a) Random selected dataset distribution. (b) Self determined dataset without class absence.

	0	1	2	3	4	5	6	7	8	9
all	8.40%	6.50%	8.60%	6.50%	3.90%	4.90%	13.00%	21.60%	15.50%	11.10%
P1	4.60%	7.20%	10.00%	5.00%	5.00%	1.80%	21.00%	15.20%	13.00%	17.20%
P2	12.20%	5.80%	7.20%	8.00%	2.80%	8.00%	5.00%	28.00%	18.00%	5.00%

(b)

all	8.00%	9.00%	21.50%	7.50%	3.50%	3.50%	8.50%	11.80%	6.50%	20.20%
P1	5.00%	4.00%	25.00%	15.00%	0.00%	7.00%	0.00%	19.20%	13.00%	11.80%
P2	11.00%	14.00%	18.00%	0.00%	7.00%	0.00%	17.00%	4.40%	0.00%	28.60%

(c)

40% 20% 0%

Fig. 7. Classes distribution of MNIST for two participants. (b) Self determined dataset without class absence. (c) Self determined dataset with class absence.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
all	8.60%	6.20%	8.60%	6.40%	4.80%	5.40%	7.00%	9.40%	7.80%	6.00%	6.00%	6.40%	6.80%	4.40%	6.20%
P1	4.80%	7.20%	10.00%	4.80%	6.80%	2.80%	8.80%	4.80%	12.00%	7.20%	9.60%	3.60%	12.00%	2.40%	3.20%
P2	12.40%	5.20%	7.20%	8.00%	2.80%	8.00%	5.20%	14.00%	3.60%	4.80%	2.40%	9.20%	1.60%	6.40%	9.20%

(b)

all	8.60%	6.20%	9.00%	6.40%	3.40%	4.00%	6.80%	9.40%	8.80%	9.00%	4.80%	4.60%	6.00%	6.00%	7.00%
P1	4.80%	7.20%	12.80%	4.80%	6.80%	0.00%	8.40%	4.80%	14.80%	7.20%	9.60%	0.00%	12.00%	3.60%	3.20%
P2	12.40%	5.20%	5.20%	8.00%	0.00%	8.00%	5.20%	14.00%	2.80%	10.80%	0.00%	9.20%	0.00%	8.40%	10.80%

(c)

Fig. 8. Classes distribution of Chinese MNIST for two participants. (b) Self determined dataset without class absence. (c) Self determined dataset with class absence.

Applying focal loss to federated XGBoost for multiclassification, the gradient and hessian should look like this: (14) and (15) shown at the top of this page.

Acknowledgment

The authors would like to thank the Associate Editor and the reviewers for their patient help and constructive review to improve this article.

References

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Statist.*, 2017, vol. 54, pp. 1273–1282.

[2] Q. Li et al., "A survey on federated learning systems: Vision, hype, and reality for data privacy and protection," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3347–3366, Apr. 2023.

[3] K. Bonawitz et al., "Towards federated learning at scale: System design," in *Proc. 2nd Mach. Learn. Syst.*, 2019, vol. 1, pp. 374–388.

[4] W. Y. B. Lim et al., "Federated learning in mobile edge networks: A comprehensive survey," *IEEE Commun. Surv. Tut.*, vol. 22, no. 3, pp. 2031–2063, Jul.–Sep. 2020.

[5] T. S. Brisimi, R. Chen, T. Mela, A. Olshevsky, I. C. Paschalidis, and W. Shi, "Federated learning of predictive models from federated electronic health records," *Int. J. Med. Inform.*, vol. 112, pp. 59–67, 2018.

[6] M. Duan et al., "Astraea: Self-balancing federated learning for improving classification accuracy of mobile deep learning applications," in *Proc. IEEE 37th Int. Conf. Comput. Des.*, 2019, pp. 246–254.

[7] B. Wang, A. Li, H. Li, and Y. Chen, "Graphfl: A federated learning framework for semi-supervised node classification on graphs," in *Proc. IEEE Int. Conf. Data Mining*, Orlando, FL, USA, Nov. 28, 2022, pp. 498–507.

[8] Q. Li, Z. Wen, and B. He, "Practical federated gradient boosting decision trees," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, pp. 4642–4649.

[9] W. Zheng, L. Yan, C. Gou, and F.-Y. Wang, "Federated meta-learning for fraudulent credit card detection," in *Proc. 29th Int. Conf. Int. Joint Conf. Artif. Intell.*, 2021, pp. 4654–4660.

[10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 785–794.

[11] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM-Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–19, 2019.

[12] L. Wang, S. Xu, X. Wang, and Q. Zhu, "Addressing class imbalance in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, vol. 35, pp. 10165–10173.

[13] A. Fernández, S. García, M. Galar, R. C. Prati, B. Krawczyk, and F. Herrera, *Learning From Imbalanced Data Sets*, vol. 10. Berlin, Germany: Springer-Verlag, 2018, pp. 978–973.

[14] L. Zhang, T. Zhu, P. Xiong, W. Zhou, and P. S. Yu, "A robust game-theoretical federated learning framework with joint differential privacy," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3333–3346, Apr. 2023.

[15] H. Yu, S. Yang, and S. Zhu, "Parallel restarted SGD with faster convergence and less communication: Demystifying why model averaging works for deep learning," in *Proc. AAAI Conf. Artif. Intell.*, Jul. 2019, vol. 33, pp. 5693–5700.

[16] J. Wang and G. Joshi, "Cooperative SGD: A unified framework for the design and analysis of local-update SGD algorithms," *J. Mach. Learn. Res.*, vol. 22, no. 1, pp. 9709–9758, 2021.

[17] X. Li, K. Huang, W. Yang, S. Wang, and Z. Zhang, "On the convergence of FedAvg on non-IID data," in *Proc. Int. Conf. Learn. Representations*, Addis Ababa, Ethiopia, Apr. 2020.

[18] J. Mills, J. Hu, G. Min, R. Jin, S. Zheng, and J. Wang, "Accelerating federated learning with a global biased optimiser," *IEEE Trans. Comput.*, to be published, doi: 10.1109/TC.2022.3212631.
[19] S. Reddi et al., "Adaptive federated optimization," in *Proc. Int. Conf. Learn. Representations*, May. 3, 2021.
[20] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," *Proc. J. Mach. Learn. Syst.*, vol. 2, pp. 429–450, 2020.
[21] J. Wang, Q. Liu, H. Liang, G. Joshi, and H. V. Poor, "Tackling the objective inconsistency problem in heterogeneous federated optimization," *Adv. Neural Inf. Process. Syst.,* vol. 33, pp. 7611–7623, 2020.
[22] D. A. E. Acar, Y. Zhao, R. M. Navarro, M. Mattina, P. N. Whatmough, and V. Saligrama, "Federated learning based on dynamic regularization," in *Proc. Int. Conf. Learn. Representations*, May 3-7, 2021.
[23] L. Gao, H. Fu, L. Li, Y. Chen, M. Xu, and C.-Z. Xu, "FedDC: Federated learning with non-IID data via local drift decoupling and correction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10112–10121.
[24] G. Ke et al., "Lightgbm: A highly efficient gradient boosting decision tree," *Neural Inf. Process. Syst.*, vol. 30, pp. 3146–3154, 2017.
[25] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," *Neural Inf. Process. Syst.*, vol. 31, pp. 6639–6649, Dec. 2018.
[26] M. Shin, C. Hwang, J. Kim, J. Park, M. Bennis, and S.-L. Kim, "Xor mixup: Privacy-preserving data augmentation for one-shot federated learning," in *Proc. FL-Int. Conf. Mach. Learn.'20*, Vienna, Austria, Jul. 18, 2020.
[27] W. Li, J. Chen, Z. Wang, Z. Shen, C. Ma, and X. Cui, "IFL-GAN: Improved federated learning generative adversarial network with maximum mean discrepancy model aggregation," *IEEE Trans. Neural Netw. Learn. Syst.*, to be published, doi: 10.1109/TNNLS.2022.3167482.
[28] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-IID data," in *Proc. GLOBECOM*, Abu Dhabi, UAE, Dec. 9-13, 2018.
[29] N. Yoshida, T. Nishio, M. Morikura, K. Yamamoto, and R. Yonetani, "Hybrid-FL for wireless networks: Cooperative learning mechanism using non-IID data," in *Proc. IEEE Int. Conf. Commun.*, 2020, pp. 1–7.
[30] M. Yang, X. Wang, H. Zhu, H. Wang, and H. Qian, "Federated learning with class imbalance reduction," in *Proc. Eur. Signal Process. Conf.*, 2021, pp. 2174–2178.
[31] H. Wang, Z. Kaplan, D. Niu, and B. Li, "Optimizing federated learning on non-IID data with reinforcement learning," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 1698–1707.
[32] Z. Tang et al., "Data resampling for federated learning with non-IID labels," in *Proc. FTL-Int. Joint Conf. Artif. Intell.*, Montreal, Canada, Aug. 21, 2021.
[33] Y. Huang et al., "Personalized cross-silo federated learning on non-IID data," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, pp. 7865–7873.
[34] D. Sarkar, A. Narang, and S. Rai, "Fed-focal loss for imbalanced data classification in federated learning," 2020, *arXiv:2011.06283*.
[35] T.-M. H. Hsu, H. Qi, and M. Brown, "Measuring the effects of non-identical data distribution for federated visual classification," in *Proc. FL-NeurIPS'19*, Vancouver, BC, Canada, Dec. 13, 2019.
[36] H. Wang, M. Yurochkin, Y. Sun, D. Papailiopoulos, and Y. Khazaeni, "Federated learning with matched averaging," in *Proc. Int. Conf. Learn. Representations*, Addis Ababa, Ethiopia, Apr. 2020, *arXiv:2002.06440*.
[37] K. Jones, Y. J. Ong, Y. Zhou, and N. Baracaldo, "Federated XGBoost on sample-wise non-IID data," 2022, *arXiv:2209.01340*.
[38] P. Kairouz et al., "Advances and open problems in federated learning," *Found. Trends Mach. Learn.*, vol. 14, no. 1/2, pp. 1–210, 2021.
[39] M. Duan, D. Liu, X. Chen, R. Liu, Y. Tan, and L. Liang, "Self-balancing federated learning with global imbalanced data in mobile systems," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 1, pp. 59–71, Jan. 2021.
[40] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 9268–9277.
[41] H. Li, X. Yang, Y. Li, L.-Y. Hao, and T.-L. Zhang, "Evolutionary extreme learning machine with sparse cost matrix for imbalanced learning," *Int. Soc. Automat. Trans.*, vol. 100, pp. 198–209, 2020.
[42] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 2980–2988.
[43] Prudential, *Prudential Life Insurance Assessment*, Zip Package Version, Feb. 2016. [Online]. Available: https://www.kaggle.com/c/prudential-life-insurance-assessment
[44] C. Leung, A. Law, and O. Sima, "Towards privacy-preserving collaborative gradient boosted decision trees," M. S. thesis, Dept. EECS, UC Berkeley, Berkeley, CA, 2020.
[45] X. Lian, C. Zhang, H. Zhang, C.-J. Hsieh, W. Zhang, and J. Liu, "Can decentralized algorithms outperform centralized algorithms? A case study for decentralized parallel stochastic gradient descent," *Adv. Neural Inf. Process. Syst.*, vol. 30, pp. 5336–5346, Dec. 2017.
[46] V. Malladi, Y. J. Li, M. Siddula, D. Seoand, and Y. Huang, "Decentralized aggregation design and study of federated learning," in *Proc. IEEE SmartWorld, Ubiquitous Intell. Comput., Adv. Trusted Comput., Scalable Comput. Commun., Internet People Smart City Innov.*, 2021, pp. 328–337.

Jiao Tian received the master's degree in quantitative finance and big data in business from Sydney University, Camperdown, Australia, in 2019. She is currently working toward the Ph.D. degree in computer science with the Department of Computing Technologies, Swinburne University of Technology, Melbourne, Australia.

Her current research interests include federated learning, incremental learning, and privacy preserving machine learning.

Pei-Wei Tsai (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic engineering from National Kaohsiung University of Applied Sciences, Kaohsiung, Taiwan, in 2005, 2007, and 2012, respectively.

He was one of the invited speakers in 2008 in the Global COE Program at Tohoku University, Japan and an invited speaker in 2021 International Symposium on Novel and Sustainable Technology, Taiwan. He was a visiting scholar in 2011 with the Center of Excellence in Information Assurance, King Saud University, Saudi Arabia. He was the Program Committee Co-chair of the 13th International Conference on Intelligent Information Hiding and Multimedia Signal Processing in 2017 and the Enabling and HCI Technologies (HCI) track co-chair of the 36th IEEE International Conference on Consumer Electronics in 2018. He is a technical committee of the Application-Specific CE for Smart Cities in IEEE Consumer Technology Society. He is currently a Lecturer with Swinburne University of Technology, Australia. His research interests include intelligent optimization, machine learning, and data analytics.

Kai Zhang is currently working toward the Ph.D. degree in computer science with the Department of Computing Technologies, Swinburne University of Technology, Melbourne, Australia.

His research interests include privacy preserve technology, machine learning, and blockchain.

Xinyi Cai received the master's degree in business information system from Monash University, Melbourne, Australia. She is currently working toward the Ph.D. degree in computer science with Swinburne University of Technology, Melbourne, Australia.

Her research interests include machine learning, natural language processing, Dialog systems, and out-of-distribution learning.

Hongwang Xiao received the bachelor's and master's degrees in the major of information management and information system from the Hefei University of Technology, Hefei, China, in 2011 and 2014, respectively, and the Ph.D. degree in computer science from the Swinburne University of Technology, Melbourne, Australia, in 2021.

He is currently a full-time employee with Beijing Academy of Artificial Intelligence, Beijing, China. Before his Ph.D. study, he was also employed in China Telecom and Baidu. He has authored and coauthored about ten papers in international high-quality journals or conferences. His current research interests include deep learning, computer vision, medical image analysis with deep learning, big vision model pre-training, few-shot learning, federated learning, medical data privacy, and related topics.

Ke Yu received the Ph.D. degree in computer science from the Swinburne University of Technology, Melbourne, Australia, in 2022.

He is currently a young post Ph.D. teacher with Chongqing University, Chongqing, China. His research interests include reinforcement learning, IoT, cloud computing, and edge computing.

Wenyu Zhao is currently working toward the Ph.D. degree in computer science with the Department of Computing Technologies, Swinburne University of Technology, Melbourne, Australia.

Her research interests include information retrieval, natural language processing, and personalized search.

Jinjun Chen (Fellow, IEEE) received the Ph.D. degree in information technology from the Swinburne University of Technology, Australia, in 2006, where he is a professor. He is a Deputy Director of Swinburne Data Science Research Institute. His research results have been published in more than 160 papers in international journals and conferences, including various IEEE/ACM Transactions. His research is in Data Security and Privacy, Cloud Computing, Scalable Data Processing and etc.